Scenario 2: Bad ConfigMap Rollout

Objective

Diagnose a rollout where application configuration and Kubernetes health probes no longer agree.

Trigger the Incident

Run this from the repository root against your own lab environment:

```
./scripts/chaos/break-config.sh student-portal
```

The script saves the current port, changes the ConfigMap, and restarts the student portal backend Deployment. Give Kubernetes a minute to observe the rollout failure before troubleshooting.

Situation

The student portal has reduced rollout capacity after a configuration release. Existing pods may still serve requests, but new pods do not become Ready.

Symptoms

- New pods fail to become Ready.
- Restart counts may increase while older pods still serve.

What You Know

- `student-portal-backend` has unavailable replicas and increasing container restarts.
- Kubernetes health events began after a rollout.
- The process in each new container appears to start.

Start With the Symptom

1. Compare Ready status and restarts on the old and new pods:

```
kubectl get pods -n student-portal -l app=student-portal-backend -w
```

1. List recent warning events and find the readiness-probe, liveness-probe, or BackOff event for the newest backend pod:

```
kubectl get events -n student-portal --field-selector type=Warning --sort-by=.lastTimestamp
```

Record the newest pod, first probe-event timestamp, and restart change.

Troubleshooting

1. Check the rollout and identify the unhealthy new pod.

```
kubectl get deployment student-portal-backend -n student-portal
kubectl get pods -n student-portal -l app=student-portal-backend
```

Expected output: the rollout is incomplete, and the newest pod is not Ready or is restarting while older pods remain Running.

1. Inspect the new pod.

```
kubectl describe pod <pod-name> -n student-portal
```

Expected output: events show readiness or liveness probe failures against port `4000`, often followed by `BackOff` or a container restart.

1. Read the application startup logs.

```
kubectl logs <pod-name> -n student-portal
```

Expected output: the application reports that it is listening on port `4001`. This conflicts with the probes checking port `4000`.

1. Compare the ConfigMap value with the Deployment probes.

```
kubectl get configmap student-portal-backend-config -n student-portal -o yaml
kubectl get deployment student-portal-backend -n student-portal \
  -o jsonpath='{.spec.template.spec.containers[0].readinessProbe.httpGet.port}{"\n"}'
```

Expected output: the ConfigMap contains `PORT: "4001"`, but the readiness probe returns `4000`. The port mismatch is the root cause.

Important Clues

Connect the probe failures to the application listening port and the port checked by the readiness and liveness probes.

Root Cause

Record the configuration and probe mismatch supported by your evidence before you apply the fix.

Fix

```
./scripts/chaos/break-config.sh student-portal --undo
```

The script restores the saved `PORT` value and restarts the Deployment.

Validation

```
kubectl rollout status deployment/student-portal-backend -n student-portal
kubectl get pods -n student-portal
```

Confirm every pod is Ready, restarts stop increasing, and the app works.

DevOps Lesson

Validate configuration and health-probe contracts together before rollout.

STAR Method

Situation

Summarize the partial-capacity rollout.

Task

Explain why only new pods needed investigation.

Action

Describe the event, log, ConfigMap, and probe comparison.

Result

State how readiness recovery was verified.

Natural Spoken Version

Use the matching example in DevOps STAR Scenarios after completing the lab.

Brief STAR Example

One of our internal applications stopped bringing new pods to Ready after a configuration rollout. My task was to identify why only the new pods failed. Kubernetes events and pod logs showed probe failures and an application listening on port 4001, while Kubernetes probes still checked port 4000. I restored the correct ConfigMap value and restarted the Deployment. All replicas became Ready, and configuration-to-probe validation was added to deployment checks.